

# Task 1: Selection and Research

*Task 1: one real regulated entity (US + at least one non-US jurisdiction); one specific regulatory domain; discussion of the political choices made by each jurisdiction.*

## 1.1 Regulated Entity: DBS Bank Limited

DBS Bank Limited ("DBS") is a Singapore-headquartered multinational banking group, the largest bank in Southeast Asia by total assets. DBS operates across 19 markets and is listed on the Singapore Exchange (SGX: D05).

DBS satisfies the cross-jurisdictional requirement through its documented US presence: a Los Angeles representative office, broker-dealer operations via DBS Securities (USA) LLC (FINRA-regulated), and resolution-plan filings with the Federal Reserve and FDIC under the Dodd-Frank Act. This makes DBS simultaneously subject to US federal oversight (OCC, Federal Reserve LISCC framework, CFPB consumer compliance) and MAS supervision in Singapore.

Three additional factors make DBS the strongest possible anchor for this assignment. *First*, DBS participated directly in MAS’s Veritas Consortium — the multi-institution working group that developed the FEAT (Fairness, Ethics, Accountability, Transparency) fairness assessment methodology, including the credit risk scoring use case. *Second*, DBS’s public Annual Reports and responsible AI disclosures explicitly reference model governance, fairness, and explainability as key AI programme concerns. *Third*, DBS’s retail credit book spans Singapore, Hong Kong, mainland China, and other markets, making a cross-border model-risk tool commercially realistic.

| Attribute             | Detail                                                                                                                              |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Legal name            | DBS Bank Limited                                                                                                                    |
| Headquarters          | Marina Bay Financial Centre, Singapore                                                                                              |
| Total assets (FY2024) | Approx. SGD 754 billion                                                                                                             |
| US presence           | Los Angeles representative office; DBS Securities (USA) LLC (FINRA); resolution-plan filer with Federal Reserve / FDIC              |
| Key US regulators     | Federal Reserve (FBO supervision); OCC; CFPB consumer compliance                                                                    |
| Key SG regulators     | Monetary Authority of Singapore (MAS) — prudential, conduct, and AI governance                                                      |
| AI relevance          | Veritas Consortium participant; FEAT Principles adopter; public responsible AI commitment in credit and wealth                      |
| FDIC filing           | <a href="https://www.fdic.gov/system/files/2024-07/dbs-165-1312.pdf">https://www.fdic.gov/system/files/2024-07/dbs-165-1312.pdf</a> |

## 1.2 Domain: AI Model Risk Management for Credit Decisioning

The domain is AI model risk management applied to retail credit-scoring and credit-decisioning models. This is the most instructive choice for this assignment because: (a) credit decision outputs directly affect access to financial services (high-stakes); (b) it is under explicit regulatory scrutiny in both the US and Singapore; (c) it is amenable to quantitative demonstration using open benchmarks; and (d) it is well-served by a mature toolkit — model validation

metrics (KS, Gini, PSI), explainability (SHAP, LIME), fairness (DI, DPD, EOD), and feature screening (IV/WoE) — that both OCC and MAS examiners would expect to see documented.

### 1.3 Regulatory Landscape: US versus Singapore

| Dimension              | us US — OCC 2026-13                                            | sg SG — MAS 2024/2025                                                                                  |
|------------------------|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Primary instrument     | OCC Bulletin 2026-13 (17 Apr 2026)<br>Rescinds SR 11-7 jointly | MAS AI MRM Info Paper (Dec 2024) MAS AI Guidelines CP (Nov 2025) FEAT Principles (2018) + Veritas v2.0 |
| GenAI coverage         | EXPLICITLY EXCLUDED — separate OCC/Fed/FDIC RFI forthcoming    | INCLUDED — all AI including GenAI agents and foundation models                                         |
| Fairness standard      | ECOA / Reg B — enumerated protected classes                    | MAS FEAT — outcome-based; no fixed statutory list; Veritas methodology                                 |
| DI threshold           | EEOC 4/5ths rule: $DI \geq 0.80$ (informal safe harbour)       | No fixed threshold — proportionate FEAT outcome monitoring                                             |
| Local XAI              | NOT mandated (implied in conceptual soundness)                 | REQUIRED for credit: per-applicant explanation for adverse decisions (MAS 2024 p.14)                   |
| Independent validation | De-emphasised: quality over structural independence            | Required for high-risk AI                                                                              |
| PSI retrain            | 0.25 (SR 11-7 lineage)                                         | 0.20 — more conservative; lifecycle emphasis                                                           |
| Min Gini               | 0.35 (US bank examiner expectation)                            | 0.30 (proportionality for smaller FIs)                                                                 |
| Monitoring             | Quarterly minimum, risk-tiered                                 | Continuous; lifecycle chain                                                                            |
| Adverse action notice  | CFPB Reg B §1002.9 — specific written reasons; AI not exempt   | MAS FEAT + PDPA Advisory Guidelines — meaningful explanation required                                  |
| IV/WoE documentation   | Implicit (conceptual soundness)                                | Explicit: developers must justify each feature used (MAS 2024 p.14)                                    |

### 1.4 Political Choices Behind the Divergences

#### The GenAI exclusion in OCC 2026-13

The OCC’s decision to exclude generative AI from Bulletin 2026-13 reflects a US regulatory preference for moving cautiously on novel technology. The exclusion was deliberately signalled — a separate interagency RFI is forthcoming — but its practical effect is that a US national bank deploying an LLM-assisted credit underwriting tool is not yet required to apply 2026-13 controls to it. MAS, by contrast, has explicitly extended its AI governance expectations to all AI systems, including GenAI agents, reflecting Singapore’s Smart Nation agenda and MAS’s track record as an early-mover on regulatory technology.

#### Fairness: enumerated classes versus outcome-based monitoring

US fairness law under ECOA and Regulation B enumerates specific protected characteristics, and the EEOC 4/5ths rule provides a quantitative safe harbour. Singapore’s FEAT Principles, developed through the Veritas Consortium with 31 industry members, including DBS, take an outcome-based approach: the bank monitors for discriminatory outcomes across any relevant grouping without a bright-line threshold. This gives MAS supervisors more flexibility but also more

ambiguity — there is no  $DI = 0.80$  safe harbour, and the bank must demonstrate proportionate oversight.

### Explainability: implied versus required

OCC 2026-13 addresses explainability through "conceptual soundness" — a model must behave in ways that can be understood and defended. This is a governance expectation, not a technical mandate. MAS's 2024 AI MRM paper goes further: it identifies credit decisioning as requiring "the most exacting standards for global and local explainability" and explicitly requires per-applicant explanations for adverse credit decisions. Under MAS, a black-box gradient boosting model without SHAP or LIME is non-compliant; under OCC 2026-13 it could still pass conceptual soundness at the portfolio level.

### IV/WoE documentation

MAS 2024 explicitly requires justification for each feature used. OCC 2026-13 implies this through conceptual soundness. In practice, IV/WoE analysis must be documented for both, but MAS's requirement is more prescriptive: a feature with  $IV < 0.02$  (useless) should not appear in a model submitted to MAS without written justification of why it was retained. The notebook implements this check explicitly.

## 1.5 Evidence of Regulatory Tensions

- **GenAI governance gap:** DBS is deploying an LLM-assisted credit underwriting pilot that needs one governance framework for Singapore (MAS: all AI in scope) and a different or absent framework for US operations (OCC 2026-13: GenAI excluded), until OCC issues separate GenAI guidance.
- **Explainability asymmetry:** the same credit model requires applicant-level SHAP waterfall explanations for MAS compliance but could in principle operate without them in the US, creating a compliance hierarchy where the more demanding jurisdiction sets the practical floor.
- **PSI threshold asymmetry:** a model at  $PSI = 0.22$  triggers mandatory retrain review under MAS (threshold 0.20) but only a monitoring alert under OCC (0.25). This creates a window where the same model is simultaneously non-compliant in Singapore and compliant in the US.
- **Fairness metric divergence:** a US examiner applies the 4/5ths rule; a MAS examiner applies FEAT outcome monitoring. The same model can simultaneously 'pass' under one regime and be under FEAT review in the other, not because of different numbers but because of different analytical philosophies.
- **IV/WoE documentation:** MAS 2024 requires per-feature written justification. OCC only implies this. A model with a weak-IV feature (e.g., Age  $IV = 0.0136$ ) requires explicit written justification under MAS that OCC would not formally require.

## 1.6 References

- OCC Bulletin 2026-13 (17 Apr 2026): <https://www.occ.treas.gov/news-issuances/bulletins/2026/bulletin-2026-13.html>
- MAS AI MRM Information Paper (Dec 2024): <https://www.mas.gov.sg/publications/monographs-or-information-paper/2024/artificial-intelligence-model-risk-management>
- MAS AI Guidelines CP (Nov 2025): <https://www.mas.gov.sg/news/media-releases/2025/mas-guidelines-for-artificial-intelligence-risk-management>
- MAS FEAT Principles (2018): <https://www.mas.gov.sg/news/media-releases/2018/mas-and-abs-launch-principles-for-fair-responsible-and-transparent-use-of-ai>
- Veritas Toolkit v2.0 (Jun 2023): <https://github.com/veritas-toolkit/>
- CFPB Reg B / ECOA AI Adverse Action Guidance: <https://www.consumerfinance.gov/about-us/newsroom/cfpb-issues-guidance-on-credit-denials-by-lenders-using-artificial-intelligence/>

- CFPB Reg B Subpart A Final Rule (Apr 2026):  
<https://www.consumerfinancialserviceslawmonitor.com/2026/04/cfpb-finalizes-regulation-b-subpart-a-rule-largely-as-proposed/>
- DBS Annual Report 2024: <https://www.dbs.com/investor/annual-report/index.html>
- FDIC DBS Resolution Plan: <https://www.fdic.gov/system/files/2024-07/dbs-165-1312.pdf>
- arXiv: Information-Theoretic Framework for Credit Risk Modeling (WoE/IV as KL divergence, 2025):  
<https://arxiv.org/html/2408.03497v3>
- PMC: Novel Framework for Transparency in Credit Scoring (LightGBM + SHAP, 2024):  
<https://pmc.ncbi.nlm.nih.gov/articles/PMC11318906/>